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THE FATE OF THE NUMERAL '4' IN GERMANIC (1)

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0. Introductory

The development of the numeral '4' in the early Germanic languages presents several interesting features. Indeed, it is no exaggeration to say that everything about it is problematic in one or other of the Old Germanic dialects - phonology, morphology, and syntax. Aside from this, the cardinal '4' holds a pivotal position in the development of the inflection of the lower cardinals in Germanic.

Yet, our standard reference works do less than justice to this complex state of affairs. To take an admittedly somewhat extreme (though not atypical) example, Bethge's discussion of the numeral in the joint compilation under the editorship of Ferdinand Dieter is the following (Bethge 1900:561):

idg. **qetuōr*: **qetur-*, vgl. got. *fidwōr* 'vier': *fidur-dōgs* 'viertägig'. Die alte konsonantische flexion im germ. stark umgebildet; schwierige lauterscheinungen (z.b. an. *fiogor* neutr.). Über das germ. *f* vgl. §112, anm. 2.

(Bethge there explains it as being on the analogy of the following cardinal *fimf* '5'.)

There must, one feels, be more to say.

Clearly, the reason I wish to make the attempt is the belief that a satisfactory account of the history of '4' in Germanic is not to be found in any one place. Further, not all the literature is conveniently listed anywhere. I have tried to notice all studies dealing with the Old Germanic languages; this explains references which are rather more copious than would otherwise be desirable. The aim of the present contribution is threefold: to present what I think has already been established (though by no means all of it codified in our standard works); to indicate remaining problems (not all of which have been properly recognized); and to suggest possible solutions to them.

1. The cardinal

1.0. The Indo-European background

The Indo-European cardinals from '1' to '19' were syntactically adjectives; like other adjectives, they could be used substantivally. Whereas

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the cardinals from '5' to '19' were indeclinable, '1' to '4' were inflected for case and gender, and thus displayed the 'Motionsfähigkeit' which is the morphological defining characteristic of the (Proto-)Indo-European adjective. ('20' and above were substantives, with the things counted taking the genitive.) The cardinal '4' was an *r*-stem plurale tantum: **k^wetuōres* m., **k^wētes(o)res(?)* f., **k^wetuōr^ς* nt.

For the *-o- of PIE **k^wetuōres*, compare WGk. τέτορες m./f., with loss of *-u-, and Arm. չ'ork'; OInd. *catvāras* has Brugmann's Law.

The special feminine form is found only in Indo-Iranian and Celtic, having been lost in all the other languages. Consequently, it need not concern us here.

The Proto-Indo-European inflection is reflected most truly in Indo-Iranian, the only branch to preserve a full set of forms. The Old Indic paradigm is as follows:

	f.	m.	n.
nom.		<i>catvāras</i>	
acc.	<i>cātasras</i>	<i>catūras</i>	<i>catvāri</i>
gen.	<i>catasṛñām</i>		<i>caturñām</i>
abl./dat.	<i>catasṛbhyas</i>		<i>catūrbhyas</i>
instr.	<i>catasṛbhis</i>		<i>catūrbhis</i>
loc.	<i>catasṛṣu</i>		<i>catūṛṣu</i>

(Szemerényi 1980:205 suggests the following skeleton Indo-European paradigm: **k^wetuores*, **k^weturns*, **k^weturōm*, **k^weturso*.)

1.1. The initial consonantism

The most obvious discrepancy between Germanic and Indo-European is in the initial consonantism. The regular outcome of PIE $*k^w$ - in Germanic should be $*h^w$ -, producing, for example, Go. + *widwōr*, OE *+hweower* (cf. IE $*k^w\delta teros$, Go. *waþar*, OE *hwæper*). Instead, we find Go. *fidwōr*, ON *fjðrir*, OE *feower*, OFr. *fiuwer*, OS *fiuuuar*, OHG *feor*, *fior*, with initial *f*-; this is also found in the combining-form and the ordinal. The discrepancy may be the result of a 'minor' sound-change process (such as a sporadic assimilation), or some form of contamination, or borrowing. Not surprisingly, in a field which has been studied as long as has Germanic, all three have been suggested.

Development of a labiovelar to a plain labial when there is another labial in the word has been held to account for examples like Go. *fidwōr*, and *wulfs* 'wolf' (< PIE $*u_l k^w os$), *fimf* '5' (< PIE $*p\epsilon nk^w e$), *wairpan* 'throw' (< PIE $*u\epsilon rg^w$ - 'twist, throw'); this was first suggested by Kluge 1886:560. But, in other examples which must be instances of the same phenomenon, there is no other labial in the word, e.g. Go. *bidjan* 'request, pray' (< PIE $*g^{wh} edh-i\alpha$ -, cf. Seebold 1967:112-113, Cowgill 1980:§10), Gmc. $*banan$ - 'death; killer' (to PIE $*g^{wh} en$ - 'strike, kill', cf. Seebold 1980:477). Moreover, some words which do contain a labial, keep their labiovelar, e.g. Go. *qiman*, OHG *qeman*. It seems more likely, therefore, that the development - if it belongs within the realm of phonology - is a spontaneous tendency of the labiovelars. Compare Martinet 1972 for a stimulating discussion.

Contamination of the initial of IE $*k^w etu\alpha$ - on that of IE $*p\epsilon nk^w e$ '5' was the solution suggested by Zupitza in his study of 1896 (7). (Similarly, the reflex in *wulfs* has been attributed to contamination with PIE $*u_l p$ - 'fox', and so on.)

Bennett has argued vigorously for a non-phonological solution to the problem (1969, with references to some of the earlier literature). He suggests that 'they are due partly to borrowing [from 'p-Celtic' -

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PVS] (which is by nature sporadic, inconsistent, and unpredictable), and partly to contamination', 1969:243.

In this situation where the all-important sound-correspondences simply do not work, it is difficult to determine exactly what process has produced the attested forms. In my view, the nature of the material suggests a 'minor regularity'; in every case we have the development of a labiovelar to the corresponding labial articulation. For this reason, I strongly favour the analysis of Martinet 1972. A case-by-case explanation by contamination, though it may hit the mark in the individual instance, is certainly atomistic; it excludes a priori the possibility of capturing any 'significant generalization'. The loan hypothesis is weakened by the absence of other pan-Germanic Celtic loans, other than the well-known cultural loans. In my view, it does not help to argue that other everyday words might have been loaned but are phonologically indistinguishable from the native stock. For these 'silent witnesses' no special hypothesis would be necessary anyway.

1.2. *The origin of the ending of the Gothic and West Germanic nominative-accusative forms*

1.2.1. Taken in isolation, OInd. nt. (pl.) *catvāri*, could, like the masculine *catvāras*, be regarded as having the long vowel by Brugmann's Law. However, we know that Indo-European had a class of 'endingless' neuter plurals with lengthened grade vowel. Further, how are we to explain the long vowel of Go. *fidwōr* if we do not have an Indo-European source for it? (Compare Meillet 1929:171.)

It follows, therefore, that the *-i* ending of Old Indic is secondary. The form *catvāri* is most straightforwardly explained as having been re-characterized with the regular Old Indic neuter plural ending *-i*. Unlike the fossilized numeral of Latin (*quattuor*) and Gothic, the Old

Indic form is part of a paradigm which opposes masculine, feminine and neuter genders.

The Gothic form *fidwōr* can ONLY have obtained its long vowel from the Indo-European neuter form $*k^w etuōr$ - there is no possible analogical source. It clearly follows from this that the regular reflex in Gothic of an Indo-European final sequence $*-ōr\#$ was $-ōr\#$. (The corollary is, of course, that the vowel retained its length in this position in Proto-Germanic.)

REMARK. Among other scholars who have derived Go. *fidwōr* from the Indo-European neuter form number Scherer 1868:448, Johannes Schmidt 1889:191-192 (they both reconstruct $+*k^w etuōri$ - from earlier $*k^w etuōr$ according to Schmidt 192), Brugmann 1890:53 fnl ($+*k^w etuōr-ə$).

It seems to me that the ending of Lat. *quattuor* also is best explained as the reflex of PIE $*k^w etuōr$ - disregarding the stem-form, which is a problem on any reckoning.

1.2.2. Strange to say, what may be fairly described as the standard view derives Go. *fidwōr* from a pre-form $+*k^w etuōres$. (Compare van Helten 1905-1906:96, Wright 1910:§89 note, Kieckers 1928:175, Krahe 1961:36-37, Krahe-Seebold 1967:116, Krause 1968:§171, Szemerényi 1980:205.)

The form $+*k^w etuōres$ is posited because of the mistaken view that, in an original final syllable, the long vowel would have undergone shortening. In its protected position in a medial syllable, the $-ō-$ supposedly retains its length. (The loss of the ending by the laws of final syllables is held responsible for the numeral becoming flectionless in early Germanic, cf. below §§1.5.0., 1.6.)

Where an attempt is made to justify the pre-form $+*k^w etuōres$, it is suggested that it is the result of the levelling of the vocalism of the neuter form into the masculine, replacing the original short $-ō-$ (so

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Szemerényi 1980:205). But there is no motivation for this; the whole assumption is ad hoc (compare the criticism of Meillet 1929:171).

The crucial objection, however, is that a pre-Germanic sequence **-ōres* would yield Go. *-ōrs*, according to the rule that pre-Germanic **-s#* appears word-finally after heavy syllables as Gothic *-s*, but is lost after light syllables (as in Go. *anþar* 'other' < pre-Gmc. **ānteros*, cf. Braune-Ebbinghaus 1981:\$78 A.2). Compare *hōrs* 'adulterer' from pre-Gmc. **kāros* (both pre-Germanic **e* and **o* are lost in Gothic final syllables). So, from *þ*^wetuōres*, we would expect Go. *þfidwōrs* (so already Boer 1921:232).

Thus, in addition to positing an incorrect pre-form, the 'standard view' is unworkable on phonological grounds.

REMARK. It must be mentioned that Krahe-Seebold (1967:61-62) claim that it is not the weight of the preceding syllable which determines the reflex of pre-Gmc. **-r^e/os* in Gothic, citing *stiur*, with *-r* after a heavy syllable, as a counter-example to the above view. Instead, they think that the relevant factor is whether the word in question is an adjective or not:

die Adjektive mit stammauslautenden *r* scheinen das *s* nicht zu assimilieren, vgl. etwa *skēirs* 'klar', *swērs* 'geehrt, gewichtig'; auch *hōrs* 'Ehebrecher' ist wohl so aufzufassen ('der Hurerische'). Die Ansicht, daß *-s* nur nach kurzem Vokal assimiliert wurde, ist wegen *stiur* 'Kalb, junger Stier' weniger wahrscheinlich.

As a purely schematic classification, this seems to work, even though it cannot be demonstrated that *hōrs* is a substantivized adjective. But there is no phonological basis at all for supposing that the mere fact that a word is an adjective rather than a noun will be responsible for different treatments of the nominative singular **-s*. It seems rather to

be chance that all but one of the heavy-syllabled words with stem-final *r* are adjectives. The nom. sg. *stiur* (only Neh 5/18, from the palimpsest Codex Ambrosianus D) remains the puzzle it always has been (despite Schulze 1934:483, 496). That the word is a masculine in Gothic, is indicated by its other attestations as 'the fatted calf', *stiur pana alidan*, in the story of the Prodigal Son: L 15/23, 27, 30. Such words as *akrs* can be viewed as retaining *-s* after syllabic *r*, or, if one does not choose to recognize the existence of syllabic consonants in Gothic, after a heavy syllable.

1.2.3. In West Germanic, OE *feower*, OFr. *fiuwer*, and OS *fiuuuar* can only be reconciled under a West-Germanic pre-form with the ending **-ar* (the Old High German form has been rebuilt, see below §1.3.2.). Economy requires that we do not separate these forms from Go. *fidwōr* without good cause. These Ingvaeonic forms thus indicate that the Proto-Germanic sequence **-ōr* was shortened to **-ar* in West Germanic.

REMARK. Shortening of a Proto-Germanic sequence **-ṽr#* in West Germanic is also indicated by the Old High German *r*-stem nominative singular in *-er*. There is no motivation for replacing the ending *+ēr* which would be expected under the rule that earlier long vowels retain their length in covered final position in Old High German, but are shortened in absolute final position (cf. Braune 1876:153, Paul 1877:332). The shortening of PGmc. **-ṽr#* sequences must therefore precede the later pre-Old High German shortening to which this rule applies.

WGmc. **-ar* > **-ær* in Ingvaeonic (of which the *-ar* of OS *fiuuuar* is a spelling, cf. Klein 1977:331-549, esp. 448-450); weakening of **-ær* > *-er* produces the endings of the Old English and Old Frisian forms.

On the North Germanic forms, see below §1.4.

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1.3. *The development of the medial consonantism in West Germanic*

1.3.0. The Proto-Germanic medial cluster $*-\delta w-$ (< PIE $*-tu-$ by Verner's Law) is retained in Go. *fidwōr*. The relation between this reflex and the forms of North and West Germanic has been the subject of dispute. In my view, the problems of both the North and the West Germanic correspondents were largely solved, departing from pre-forms with a dental, as in Gothic, already at the beginning of this century (see §§1.3.2., and 1.4.1.-1.4.3.). Regrettably, these explanations have often been unjustly dismissed.

1.3.1. In so far as the phonological development of the West Germanic forms of the numeral is discussed at all in the recent literature, the explanation generally given is one first tentatively put forward by Kluge in a footnote of his dissertation, 1879a:102-103 [footnote]. This presumes a Germanic pre-form $*feywōr$, usually derived from a pre-Germanic $*k^w ek^w-$, supposedly assimilated from PIE $*k^w etu-$.

I think that the reconstruction of this pre-form is untenable. In the first place, the Germanic data at our disposal only evidence a pre-form with initial $*p-$ ($*f-$), cf. §1.1. above. This means that the assimilation would have to have taken place before the replacement of $*k^w-$.

Yet the proposed assimilation cannot have affected all parts of the (pre-) Germanic speech community, because we need $*fēδw-$ to account for the Gothic form. (A dental is also needed for the pan-Germanic combining-form, cf. §2.) The chronological and dialectal complications which this theory entails are far from plausible. For criticism of Kluge's suggestion, cf. also Schmidt 1880:123-124.

REMARK 1. Kluge himself confessed of his explanation: 'wenn die got. Formen mit *d* fehlten, würde sie mehr Wahrscheinlichkeit haben' (1879a:102-103) - and this the form, which is most clearly the direct

continuant of the Proto-Indo-European etymon! We may note that Kluge modestly followed the exposition of his explanation with the words: 'Ich stelle meine Annahme nur in der Hoffnung auf, dass sie bald durch eine schlichtere, einfachere ersetzt werden möge' (1879a: 102-103). Unfortunately, this did not happen.

What is more, we cannot even arrive at the West Germanic forms from the proposed starting-point. It is generally agreed that a PGmc. **feywōr* would develop to either WGmc. **feyar* or **fewar* (cf. most recently Cubbin 1979, with references), yet Old Saxon *fiuuuar*, with its triple <u>, must descend from a precursor with the sequence **-uw-* < **-ww-* (cf. Gallée-Lochner 1910: §106 A.2). Likewise, the Old English and Old Frisian forms, *feower* and *fiuwer*, are most straightforwardly derived from such a pre-form.

Old English forms, such as Rushworth 2 <feower>, require a preceding **-ē-*. We can thus establish a sequence **-ēww-* for West Germanic (besides having no etymological justification, **-ēww-* was a non-occurring sequence of sounds). On the Old English development of **-ēww-*, cf. Campbell 1959: §§120(2), 275. (We may note that, in keeping with his view, Kluge [1879b:395 fn1] denied the length of the diphthong in OE <feower>; this would necessitate deriving the short diphthong by breaking.) On the numeral in Old Frisian, cf. van Helten 1890: §§36, 234, 236; Steller 1928: §27 A.2 [p.22 base] - Steller's treatment of the numeral lacks consistency, however; compare the contents of §§9.2, 20.b, 27 A.2 [p.22 base].

REMARK 2. The *i*-vocalism of the Old Frisian and Old Saxon forms is clearly secondary. Siebs (1901:1233) operates with a raising of WGmc. **eu* to **iu* before *w* (so also Steller 1928: §20.b, Sjölin 1969:22), which would account for the Old Frisian form. This development seems doubtful, however, in view of such forms as *treuwe* in the Rürstringer Recht, cf. Siebs 1901:1236-1237. What is more, earlier *e*-vocalism

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would appear to be reflected in the (secondary) combining-form *fīār-* (cf. §2).

It seems conceivable to me that the *i*-inflection of the numerals has been responsible for this replacement of **e* by *i* in these forms of '4'. The *-a-* of the medial syllable of inflected forms may have assimilated to *-i-*, which will have changed *-eu-* to *-iu-* in the initial syllable.

Kluge, in a slightly later contribution (1882:517-519, essentially repeated 1913:252), van Helten (1905-1906:84) and Hamp (1976:1) have all derived the same supposed PGmc. pre-form +**feɣwōr* as Kluge originally advocated, by other equally, or even more implausible means.

1.3.2. The Ingvaeonic evidence thus assures **fewwar* as the West Germanic pre-form. Comparison with PGmc. **feðwōr* argues strongly for a regular development *-ðw- > *-ww- in West Germanic. This was hinted at by Leskien 1876:155-156, but first stated clearly by Hartmann 1900:304, apparently independently.

Jansson (1936:285) took issue with Hartmann over OHG *fior*, which he did not believe could come from **feww-*. Instead, he preferred a development: WGmc. -**feuōr* (< +**feɣwōr*, as discussed above) > *feor* > *fior*, by loss of intervocalic -u-. However, loss of intervocalic -w- occurs sporadically after LONG vowel in Old High German, but not after short. One COULD argue, therefore, for a derivation of OHG *fior* from WGmc. **fewwar* via loss of -w- after (reflex of) the diphthong -eu- and subsequent contraction of **feuär*. In my view, however, the Old High German cardinal is best explained as being secondarily extracted from the ordinal, OHG *feordo*, *fiordo* (< PGmc. **feurpan-*, cf. §3, below). This also explains the *eo*-vocalism of early forms (see *Althochdeutsches Wörterbuch* s.v. *fior*), cf. Braune-Eggers 1975:§48 A.5; according to their §30 A.2 (cf. §113.2), we might

expect OHG *+fiuwar* as the Old High German reflex of WGmc. **feubar*. (The numeral later undergoes independent contraction in the other West Germanic languages, losing the reflexes of WGmc. **-wa-*, thus IOE *fēor*, OFr. *fior* (**feor*, cf. §2), OS *fior*, in so far as this last is not to be identified with the Old High German form.)

Jansson continues: 'På samma sätt [som ght. *fior* - PVS] kan fsachs. *fior* uppfattas', and goes on (like Kluge, above §1.3.1.) to claim that OE *feower* has breaking, and mysteriously adds: 'På samma sätt som ags. *feower* måste väl även ffris. *fiuwer*, fsachs. *fiuwer* förklaras' (1936:285), despite the fact that these dialects have no soundchange equivalent to Old English breaking of *e* before *w*!

1.3.3. Not only does confrontation of the West Germanic and Proto-Germanic pre-forms suggest the correspondence WGmc. **-ww-* : PGmc. **-ðw-* is regular (though see below, §1.3.4.), but there is also the supporting evidence of the parallel development of the sequence **-zw-*, which is generally acknowledged to yield WGmc. **-ww-* in the oblique cases of the second person plural pronoun. This development was first explicitly proposed by Kluge 1908-1909:65, cf. also Trautmann 1912:107; the implicit equation was already made by Kögel 1884:536-537. Accordingly, corresponding to Go. acc., dat. *izwis*, we have OE *ēow*, OFr. *iu*, OS *iu*, OHG *iu*; to Go. gen. *izwara*, OE *ēower*, OFr. *iuwer*, OS *euwar*, OHG *iuwēr* (with secondary ending).

Thus, the two Proto-Germanic voiced dental spirants are assimilated to a following **-w-* in West Germanic. This soundchange must be very early, for it precedes both the occlusion of PGmc. **ð*, and the rhotacism of PGmc **z*. It is ironic that Hartmann's soundlaw **ðw- > *-ww-* was set up, and rejected, for Old High German, when it is the Ingvaemonic forms which are most elegantly explained by it. To be fair, however, it must be pointed out that Hartmann was aware of the difficulties connected with the cardinal '4' in Old High German, (1900:304):

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zw und ðw scheinen westg. zu ww assimiliert worden zu sein; got. *izwara* ahd. *euuuēr*, *iuuuēr*, got *fidwōr*, *fidur* ahd. *uëor*, *fior* (ausgleichung aus *fiuuuor* und *fidur*), doch bieten namentlich die formen des zahlworts grosse schwierigkeiten.

1.3.4. The above development of PGmc. **-ðw-* > WGmc. **-ww-* in medial position is only evidenced in the cardinal '4', though a number of other words provided the requisite environment. These apparent counter-examples are all *wa-* or *wō-*stem nouns (or their derivatives); there do not seem to have been any *wa/ō-*stem adjectives with root-final **ð*. The examples I have noted are the following: the *wa-*stems **gaiðwa-* nt. 'lack', **skaðwa-* nt./masc. 'shade' (in Old English also a feminine), and, possibly, **kwiðwa-* nt. 'cud; mastic'; the *wō-*stems **mēðwō-* 'meadow', **baðwō-* 'war, battle'. To the latter, the precursors of OE *frēod* 'good-will; peace', *hīd* 'hide (of land)' are possibly to be added, if the etymologies proposed by Bammesberger (1979: 54-55, 74-76) are accepted.

This material will bear some winnowing. Firstly, the word for 'cud' is probably an original *u-*stem. Only Old English has *wa-*stem forms, and according to Ross 1954:94-96 these are analogical. PGmc. **frijaðwō* and **hiwiðwa/ō* have become simple *a-* (*ō-*) stems in Old English. If the rebuilding of these two could legitimately be placed before the soundchange, they could be removed as counter-examples. **gaiðwa-* (cf. Go. acc. pl. *gaidwa* Col 1/24) might possibly be explained in the same way: Old English *gād* (1 x *gæd* GenB 236, cf. Bammesberger 1979: 58) is found only in poetry and is attested only in the nominative singular; Old Saxon only has the word as the second element of a compound, gen. pl. *-gēdeono* Hel M 4331 (C *-gēdono*), apparently a *jō-*stem form, cf. Ilkow 1968:292; Old Frisian has lost the *wa-* and *wō-*stem categories, the relevant lexemes having become strong nouns. One cannot be certain of the chronology, however; in fact the change of inflectional class of these three nouns can perhaps be most neatly

explained as a result of the (post-West Germanic) loss of *-u* after heavy syllables. Besides, acceptance of this explanation would still leave the other items to be accounted for.

One might suppose that sufficient case-forms regularly developed the sequence **-ǫu(-)* before the assimilation took place, and consequently did not undergo it, to cause the assimilation-product to have been levelled out of the paradigm. This does not seem to work, however. Assuming that the assimilations of **zw* and **ǫw* to **ww* were simultaneous (indeed the same change, §1.3.3.), they must have taken place before the rhotacism of medial **-z-*, and this preceded the loss of final **-a#* (as shown by, e.g. OE *dēor*, OHG *tior*, beside Go. *dīus**, < PGmc. **deuzaⁿ*). This means that **-ǫwa(z)#*-sequences will have been still intact at the time of the assimilation, so the only case-forms with *-ǫu(-)* that can be produced are those which lost *w* before following *u*: The masc./nt. dat. pl. **-um*, and the nt. nom./acc. pl. ending **-u* (and possibly an instr. sg. in **-u*) for the *wa*-stems, and the nom. sg. **-u* in the *wō*-stems. This seems an inadequate basis for analogical elimination of the change.

I must confess, therefore, that I am not wholly sure how these exceptions are to be explained, though I am sure that the development posited for '4' is phonological. At present, I will only offer the following tentative suggestions as to why the *wa*- and *wō*-stem nouns should have escaped the change. It is possible that the development of '4' is connected with the fact that it had a final *-r#*, which might have triggered some kind of 'similatory' change of the preceding **-ǫ-* to **-z-*, which then regularly assimilated to the following *-w-*. Alternatively, it is possible that, in the heavy-syllabled forms, the effects of Sievers' Law as it applies to *u* and *uw* played a role in inhibiting the change in the substantives (the isolated light-syllabled form **feǫ-wōr*, on the other hand, was unable to escape it).

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1.3.5. It will be most convenient to consider the medial consonantism of the North Germanic forms in connection with the strong-adjective inflection which is found there, see §1.4. On 'Crimean Gothic' *fyder*, see §1.6.4.3.

1.4. *The North Germanic situation*

1.4.0. As remarked above (§§1.2.3., 1.3.5.), the phonological questions connected with the North Germanic data are best treated in the context of the strong-adjective inflection which is found in this branch of Germanic. This is because, under the most satisfactory interpretation, the individual case-forms are the conditioning environments for the postulated changes. This explanation was worked out in an exchange between Noreen and Kock at the turn of the century: Noreen 1890:317-318, 1892:§106a, Kock 1895:125-134, Noreen 1897:39-43, Kock 1898:252-254, repeated 1917:269-270. (The elements of the solution were already given in embryo by Bugge 1873-1895 [1873-1878]:61-63.) It is presented in Noreen 1903:§§251 and 282 (also 1923:§§259 and 292), 1904:§§228 and 244.3, 1913:§§61c, 85.9a, 218; it was endorsed by van Helten 1905-1906:97. It is usually represented in the standard works on North Germanic (e.g. Boer 1920:§285, Heusler 1932:cf. §296, Hellquist 1939: s.v. *fyrá*, and Brøndum-Nielsen 1928-1973:I.181,II.28). Outside the field of North Germanic, this explanation has been largely disregarded.

REMARK. It will be noted that I am in almost total disagreement with the views of Jansson 1936; this is one of the reasons the present study came to be written. I am also unable to accept the suggestions of Harding 1942, 1944.

1.4.1. I give below the regular Old Icelandic paradigm of '4' and, for comparison, that of the plural of *gamall* 'old':

f.	m.	nt.		f.	m.	nt.
	<i>fjðrir</i>		nom.		<i>gamlir</i>	
<i>fjðrar</i>		<i>fjögur</i>	acc.	<i>gamlar</i>		<i>gömul</i>
	<i>fjðra</i>		gen.		<i>gamla</i>	
	<i>fjögurra</i>		dat.		<i>gamalla</i>	
	<i>fjðrum</i>				<i>gömlum</i>	

In '4', we clearly have a synchronic stem-alternation of *fjðr-* pre-vocally, and *fjögur-* elsewhere; a short vowel plus *-gu-* alternates with a long vowel minus *-gu-*. Moreover, both *-jð-* and *-jǫ-* can be interpreted as breaking diphthongs (as was done by Kock 1895:126), the latter a lengthening of the former. The lengthening can in turn be interpreted as compensatory upon the loss of a following consonantal element, which would in turn correspond to the actually occurring *-g-*. (Synchronically, *-gu-* and length are in complementary distribution, as pointed out above.) Kock further compared the alternation of stem-forms with those that occur in the inflection of *gamall* (1895:127):

Aus den angeführten paradigmten ergibt sich also, dass sich bei dem zahlwort 'vier' *fjó*-formen in a l l e n den casus und n u r in solchen casus finden, wo *gamall* synkopierte formen hat, während das zahlwort mit *fjög-* in a l l e n den casus und n u r in solchen casus anfängt, wo *gamall* unsynkopiert geblieben ist.

Diese übereinstimmung ist allzu gross, um zufällig zu sein.

Kock rightly stressed the original unity of the parent paradigm, 1895:126:

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Es ist selbstverständlich, dass *fiōrir*, wenn irgendwie möglich, in einer solchen weise zu erklären ist, dass es nicht von den übrigen casus desselben wortes (*fiogur*, *fiogurra*) getrennt wird.

Noreen, for his part, quite properly insisted that the same (dental) consonantism should be posited as is reflected in Gothic (cf. particularly 1897:41) - this was later accepted by Kock 1898:253.

1.4.2. The development postulated was as follows. Underlying the attested North Germanic forms was a stem **feður-*, furnished with the ordinary strong adjective inflection (Noreen 1897:41). The stem-form **feður-* is immediately reminiscent of the Old Indic zero-grade stem-form *catūr-* (above §1.0.; and had indeed been equated with it by Bugge 1873-1895 [1873-1878]:62) - see further, §1.6. below. This gives the following (late) Proto-North Germanic paradigm:

	f.	m.	nt.
nom.	<i>feður-ōR</i>	<i>feður-ēR</i>	<i>feður-u</i>
acc.		<i>feður-an(n)</i>	
gen.		<i>feður-ēRō</i>	
dat.		<i>feður-umR</i>	

1.4.3. The **-u-* in the syllable following the **-e-* caused breaking. This change, syncope, and the subsequent vowel shortenings will have led to the paradigm below. (On the representation of the breaking diphthong, see §1.4.5.)

	f.	m.	nt.
nom.	<i>feudraR</i>	<i>feudriR</i>	<i>feudur</i>
acc.		<i>feudra</i>	
gen.		<i>feudurRa</i>	
dat.		<i>feudrum</i>	

Next intervene two sound laws adduced by Noreen: $*-\ddot{o}- > \emptyset / r$, and $*-\ddot{o}- > -\gamma- / u$ (1897:39-43, esp. 40, cf. also 1904:§§244.3 and 228, 1923:§§292 and 259). Their ordering relative to each other cannot be determined, but they must both come later than the syncope of *u* after light syllables (so Noreen 1897:41), cf. further §1.4.6. With compensatory lengthening consequent on the loss of $-\ddot{o}-$, we arrive at the attested paradigm, given above §1.4.1.

1.4.4. More needs to be said on the vocalism of the stem-syllable of the North Germanic forms. The West Norse by-forms ONw. nt. pl. *fjugur*, gen. pl. *fjugurra* (cf. Noreen 1923:§448), and the constant <ju>-spelling of East Norse (cf. Noreen 1904:§483) must be accounted for.

Let us consider them in relation to the two rival theories of breaking. According to the 'breaking-umlaut theory', *a*- and *u*-breaking produced in the first instance a single reflex (which may be written *ia*). Later, those *ia* reflexes which were the result of *u*-breaking developed to *iɔ*, by the independent process of labial mutation.

The alternative view (the so-called 'epenthesis theory', cf. Benediktsson 1963:431, 1982:39) regards *a*- and *u*-breaking as having produced reflexes which differed from one another phonetically from the very beginning. (I might just point out that, under this second view, the products of *a*- and *u*-breaking cannot have been contrastive until the loss of *u*; until this occurred, any phonetic distinction between the

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two will have been phonemically neutralized. The two theories thus diverge in their accounts of how the product of *u*-breaking is incorporated into the phonological system.)

As Benediktsson has pointed out (1982:40), the 'breaking-umlaut theory':

implies that, subject to possible exceptions which could be explained as phonologically regular deviations, the vowel of the combination ultimately resulting from the breaking of *e* before an *u* should be identical with the *u*-umlaut vowel *ø*, and this is indeed not the case. In West Nordic there are such instances as ONorw. *þiukkr* 'thick', nt. *fiugur* 'four', where the spelling shows an unambiguous deviation from the *u*-umlaut vowel (while the regular spelling with 'io' is ambiguous, denoting either *io* or *iø*), as well as OIcel. *þiokkr*, which is confirmed by an eleventh-century rhyme (*flokki* : *þiokkva*).

Compare Benediktsson 1963:428-431, where he notes greater fluctuation in the quality of the vowel in East Norse (430).

Benediktsson - in my view rightly - considers this data as fatal to the 'breaking-umlaut theory', and continues (1982:40):

The point is that, while the vowel of the *a*-breaking cluster (*ia*) was naturally and readily identified with *a* as the only unrounded back vowel, the vowel of the *u*-breaking cluster may have been readily identifiable - and was ... in fact alternately identified - with any of the three rounded back vowels, *u*, *o*, and *ø*, since there was no tongue-height contrast in the position immediately after a *j*. Therefore, once the *a*-breaking vowel was identified with *a*, the *a*- and *u*-breaking alternation was readily susceptible of a secondary attraction to the *u*-umlaut alternation $a \rightarrow \text{ø}$ (that is, appearing as an

alternation *ja* → *jǫ*), since both alternations were operative under quite similar morphological conditions.

In the original version of the report (1980:34), the following paragraph appears at this point:

The exceptions like *þiokkr*, etc., on the other hand, are more isolated as far as morphological alternation is concerned, and therefore more likely to have the status of phonological relics and thus to be close representatives of the original *u*-breaking cluster, not identical with the *u*-umlaut result *ǫ* < *a*.

The numeral '4', with the invariant stem-form **feður*- posited above, clearly comes under the head of isolated words. The actually occurring forms fit into the pattern of attestation discussed by Benediktsson 1963. The 'epenthesis theory' of breaking can thus explain the vocalism of the various North Germanic forms of '4' (and also a wide range of other apparent aberrations) without recourse to any additional hypotheses.

1.4.5. The soundlaw whereby 'ð mellan ett konsonantiskt och ett sonantiskt *u* öfvergår til ʒ' was set up by Noreen (1897:40) to account for OSw. *iugher* 'udder', which he derived from **iuγur*, from earlier **iuður* < PNGmc. **euður(a)*. He then proceeded to apply this soundlaw to other forms, including the unsyncopated forms of the numeral '4', whereby (to adopt spellings consonant with those used by Noreen) **feður*- > **fiuður*- > **fiuγur*- (cf. already Bugge 1873-1895 [1873-1878]:62). It must be pointed out, however, that the two examples are not exact parallels (as noted by Kock 1895:130). In the former instance, we have a PGmc. diphthong **eu*, and in the latter, the *u*-breaking product of PGmc. **e*. The spellings used by Noreen obscure this important difference. (It was because of this that <eu>

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spellings were used above, §1.4.3., when reporting Noreen's explanation.) All the other examples given by Noreen (1897) have original *u*-diphthongs too. Clearly, we must widen the environment of Noreen's soundlaw so as to include instances when the **ð* is preceded by the vowel of the *u*-breaking cluster. (The retained *u* in the following syllable may well have caused a particularly high, round variant of the vowel in the breaking cluster.) This feature of the conditioning environment could also be taken as a weak indication of the correctness of the 'epenthesis theory' of breaking, above §1.4.4.

1.4.6.0. A special word is needed on the earliest North Germanic attestations of the numeral '4', the Younger Runic forms **fiakura** acc. pl. masc. and **fiakurum** dat. pl. on the Rök Stone. These diverge from the forms normally found in literary Old Norse. Benediktsson's equation neatly sums up the discrepancy (1963:431): 'Rök *fiakura*, *fiakurum* (= Olcel. **fiǫgura*, **fiǫgurum* = *fiðra*, *fiðrum*)'. It is important to ascertain how the Rök forms fit in with the explanation of the literary forms which has been elaborated above.

In the orthography of the Rök inscription, <**k**> represents /*γ*/, and <**ia**> is a spelling for the *u*-breaking diphthong in a position of neutralization before a retained *u* (cf. Benediktsson 1963:431). What needs clarification is the appearance of this allomorph of the stem-form in these case-forms.

1.4.6.1. At first sight, the Rök forms appear to make a mockery of the chronology set up above on theoretical grounds in order to derive the literary forms (cf. especially §1.4.3.). To judge by the forms **sunu** acc. sg., **karuR** nom. sg. < **garwaR*, **fiaru** < **ferwa* < **ferhwa* (and **sitir**, **nipiR** < **nipjaR*), Rök does not yet have syncope of *u* (and *i*) after light syllables. Above (§1.4.3.), the change of **ð* > **γ* /*u* *u* was placed LATER than the loss of *u* after light syllables. Yet in **fiakura** and **fiakurum** we have spellings indicating that the change has taken place

BEFORE syncope; as a result we have the consonant *-g-* in case-forms which are syncopated in the literary Norse languages. It would appear, therefore, that the literary forms with compensatory lengthening owe their length to a loss of **γ* before *r* after second syncope and not one of **ð* as supposed above. This is the development Kock tried to establish (1895:128-131). But, unlike the loss of **ð* before *r*, there seems to be no good evidence for an early Nordic loss of **γ* in this position.

On the other hand, I do not see how the distribution of allomorphs attested in literary Norse could be the result of analogical reformation on the model of the type *gamVI-/gaml-* (cf. §1.4.1.). Starting from a paradigm with a phonologically-developed unvarying stem-form **fiuγur-* (corresponding to **gamal-*), the result would surely be something like ON *+fjǫgur-/+fjǫgr-*. It would appear, therefore, that the parallel with *gamall* is genuinely phonological (as was suggested above, §1.4.1.), or, at least, that the Old Norse paradigm of **gamal-* is modelled on the type represented by **fiuγur-*.

1.4.6.2. We could remove the embarrassment, however, if we refined our concept of the syncope of *u*. It is true that the syncope of *u* after light syllables has not yet taken place in FINAL syllables in the Rök language. But, if we assume that syncope of *u* after light syllables has taken place in MEDIAL syllables, we can get round the difficulty. It is no longer necessary to posit a general Nordic loss of **γ* before *r*, for Rök now poses no problem for dating the change of **ð* to **γ* later than syncope of medial *u*.

REMARK. With reference to §1.4.3., it is interesting to note that not only the change of **ð* to **γ*, but also the loss of **ð* before *r* is evidenced in the language of Rök, cf. **þiaurikR** 'Theodoric', beside (reborrowed?) OIc. *þjóðríkr*.

1.4.6.3. It would then follow that the Rök forms **fiakura** and **fiakurum**, with the stem-alternant *fjögur-* in the wrong case-forms, are to be interpreted as the result of (an isolated) paradigmatic levelling (so already Bugge 1873-1895 [1873-1878]:62, also Noreen 1897:41; similarly Kock 1898:254, 1917:272-273). A form *fiugrum* (Schlyter 1841: 130 n27), found in one of the manuscripts of the *Vestmanna lag* (Cod. Holm. B 55, Schlyter's MS D, 1841:xxx), can be cited in support of this interpretation, though it must be pointed out that the Rök Stone (Lysing härad, western Östergötland) and Västland are not particularly close to one another. (Obviously, when we are dealing with three attestations of the same lexeme, of which two occur in a single text, it is theoretically preferable to postulate isolated analogical developments rather than an isolated phonological development which contradicts the pan-Nordic relative chronology of soundchanges.)

1.4.6.4. The theoretical assumption made above concerning the relative chronology of syncope of medial *-u-* (and *-i-*) after light syllables is not completely ad hoc. The forms (**ub**) **fatlabR** 'girt (round)' < PNGmc. **fatilōðaR* and **tualfta** (*n*-stem nt. nom. sg.) 'twelfth' < PNGmc. **twaliftō* directly attest the loss of medial *-i-* after a light syllable in the language of the Rök inscription. (Rök **tualf** 'twelve' must be analogical on the ordinal, so Grønvik 1983b:136.) It seems possible, therefore, to regard the existence of the forms **fiakura** and **fiakurum** (with the change of **ð* to γ) as providing indirect evidence for the corresponding loss of medial *-u-* after a light syllable.

I am grateful to Ottar Grønvik for drawing the above evidence to my attention and suggesting its possible implications (cf. Grønvik 1983b:136).

1.4.7. It need hardly be pointed out, that positing a pre-Germanic zero-grade form $+*k^w ek^{(w)}ur-$ (cf. §1.3.1.) does not provide a better

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account of the North Germanic forms. Labiovelars were delabialized before *u*, in Germanic (cf. PGmc. **gunþ-* 'fighting; battle', a zero-grade form to the root **g^{wh}en-* mentioned in §1.1.), if not already in Indo-European. It follows, therefore, that this pre-form would not yield **fewur*-reflexes (we would have to assume that they were created by some kind of analogy). Further, these and the regular **feȳur*-reflexes would somehow have to receive just the right repartition so as to produce the literary forms. As there is no linguistic motivation for such a development, it would have to be entirely the result of chance.

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(To be continued in *NOWELE* vol. 7)